SIGNBRIDGE

Revolutionizing Accessibility through Robotic Sign Language Production

Authors: Ala Belguith, Hedi Gouiaa, Mohamed Ayoub Ben Ayed

Supervisor: Bechir Zalila

National Engineering School of Sfax (ENIS)

May 2026

THE COMMUNICATION GAP

Motivation

According to the World Health Organization (2025/2026 updates), over 430 million people worldwide live with disabling hearing loss, a number projected to rise to 700 million by 2050

The Vision

Making communication more accessible by leveraging modern robotics and mobile technology to bridge the silence with physical, real-time sign language simulations.

The World Federation of the Deaf estimates that about 80% of deaf individuals lack access to formal education, making even basic digital text difficult to understand for many.

BARRIERS TO ACCESSIBILITY

Limited Tools

Severe lack of affordable, real-time communication devices.

Interpreters

High cost and limited availability of human interpreters.

Text Constraints

Standard text lacks the emotional nuance of signing.

Intuition

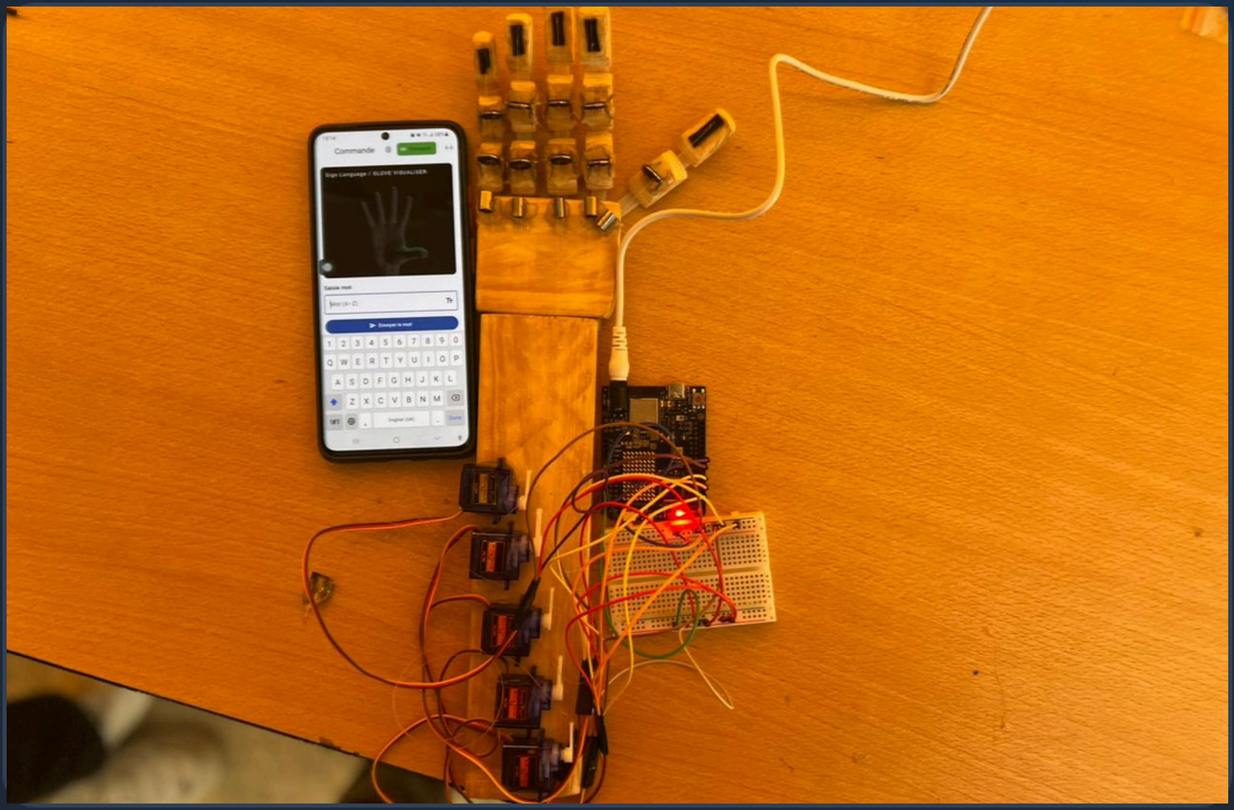
Need for a physical interface rather than just digital avatars.

THE SIGNBRIDGE SOLUTION

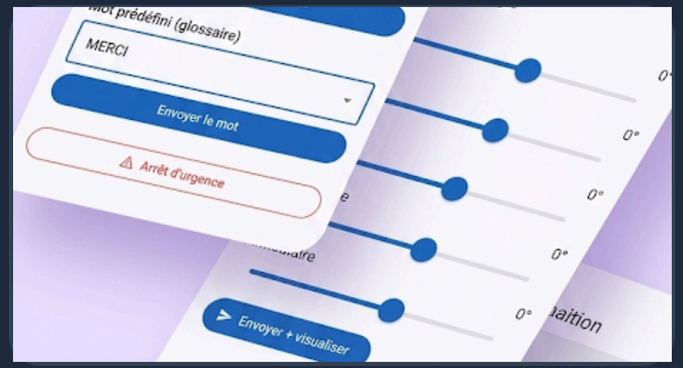
Text-to-Gesture Engine

SignBridge is an end-to-end system designed to translate text input into physical sign language gestures using a wood maded robotic hand.

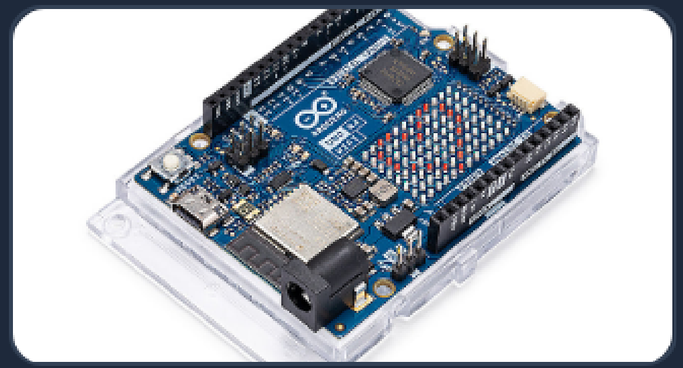
- Real-time processing and translation
- Controlled via intuitive Flutter mobile app
- Wireless operation (BLE/Wi-Fi)
- Highly accurate finger positioning



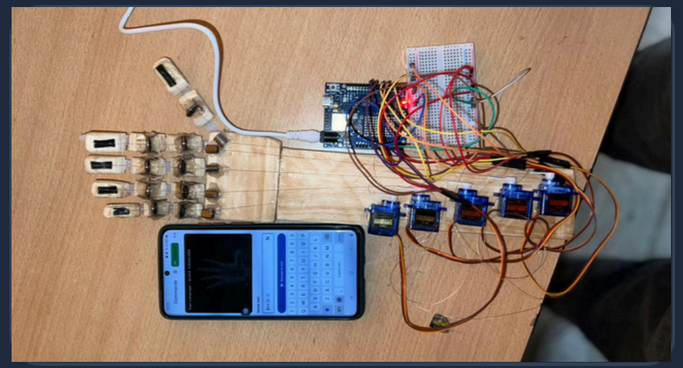
INTEGRATED SYSTEM OVERVIEW



Frontend
Smartphone with Flutter app for user interaction.



Core Logic
Arduino UNO R4-WiFi processing incoming JSON commands.

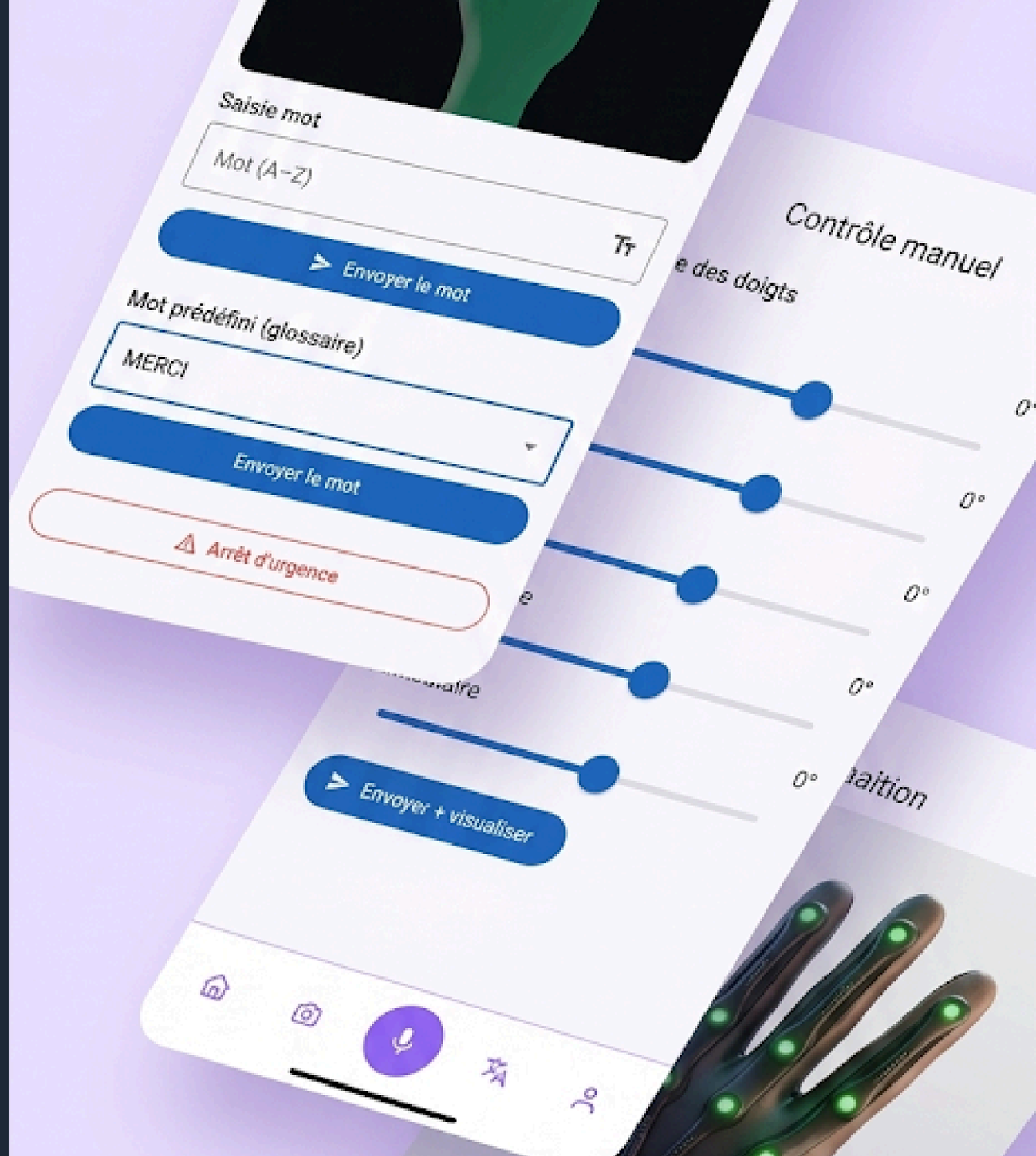


Execution
PWM-controlled servo motors actuating a wooden hand.

FLUTTER INTERFACE

The presentation layer provides a seamless user experience for controlling the robotic hand hardware.

- **Text Input:** Direct conversion of phrases
- **3D Hand Model:** Real-time digital mirroring
- **Manual Mode:** Individual finger articulation
- **Status:** Live feedback on hardware connection



DUAL-MODE COMMUNICATION

This allows the mobile app UI (User Interface) to "listen" to the connection status.

```
final StreamController<bool> _connectionController =
    StreamController<bool>.broadcast();
Stream<bool> get connectionStream => _connectionController.stream;
```

Sends Via BLE

ble_gesture_service.dart

```
if (ConnectionMode ==_mode  && anmionMode != true) {
  try {
    await _gestureChar.!ble.ble.write(bytes, false);
    return null;
  } catch (e) {
    return e;
  }
}
```

Sends Via WiFi in case of BLE Failure

socket_service.dart

```
if (_mode == ConnectionMode.wifi && _wifiSocket != null) {
  try {
    _wifiSocket!.add(bytes);
    _wifiSocket!.add([0x0a]);
    _wifiSocket;
    return true;
  } catch (e) {
    return false;
  }
}
```

THE EMBEDDED BRAIN

4 Control Layers

The Arduino UNO R4 WiFi serves as the central hub, parsing JSON packets into precise motor movements.

Hardware Components

- Arduino UNO R4 WiFi (RA4M1 ARM Cortex-M4)
- 5x TowerPro SG90 Servo Motors
- Tendon-based movement mechanism
- External 5V regulated power supply

GESTURE DATASET MAPPING

Deterministic mapping of alphabet characters to finger angles (0° - 90°).

Character	Thumb Angle	Index Angle	Middle Angle	Ring Angle	Pinky Angle
A	0°	90°	90°	90°	90°
B	90°	0°	0°	0°	0°
C	45°	45°	45°	45°	45°
D	90°	0°	90°	90°	90°

Formula used for PWM signal: *P= (angle/180 * 2000) +500*

AGILE SCRUM METHODOLOGY

Sprint 1

Connectivity & Hardware Setup

Sprint 2

UI Design & Comms Layer

Sprint 3

3D Visualization & Testing

Sprint 4

Final Prototype & Validation

PERFORMANCE METRICS

Average Latency measured at **~850 ms** (Target: < 300 ms)

ROADMAP & LIMITATIONS

Current Limitations

- One-directional (Text → Sign only)
- Fixed wrist mechanism
- Dependency on manual calibration
- Latency above real-time target

Future Work

- Bi-directional (Sign → Text) via IMU Glove
- Full wrist and elbow movement
- AI-powered predictive signing
- Sensor integration (Flex & IMU sensors)

CONCLUSION

"SignBridge successfully demonstrates that the fusion of software, hardware, and bio-inspired robotics can create tangible impacts on human accessibility."

— Project Final Summary

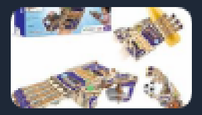
We have delivered a functional end-to-end prototype that translates digital thought into physical action, paving the way for more inclusive robotic communication.

Thank You

Questions &

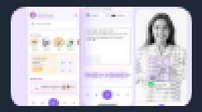
Discussion

IMAGE SOURCES



https://www.smartivity.in/cdn/shop/files/1_20_grande.jpg?v=1740515128

Source: www.smartivity.in



https://cdn.dribbble.com/userupload/5550016/file/original-228aa656e36c98210429f68c18b8bce0.png?resize=2048x1536&vertical=center

Source: dribbble.com

https://store.arduino.cc/cdn/shop/files/ABX00087_01.iso_65d9153b-9fe3-4d51-8f0a-750fcca31c5e_934x700.jpg?v=1749566714

Source: store.arduino.cc

https://kitsguru.com/cdn/shop/files/robotic-hand-kit-gripper-front.jpg?v=1765172865

Source: kitsguru.com

https://www.bunnyshell.com/images/blog2/what-is-object-storage/hero.jpg

Source: www.bunnyshell.com